

Hira Nawaz

MS Computer Science Researcher | AI for Healthcare | Medical Imaging | Trustworthy AI

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LinkedIn | GitHub | Kaggle

RESEARCH INTERESTS

Trustworthy AI for Healthcare | Medical Imaging | Multimodal Learning | Vision-Language Models | Explainable AI | Uncertainty Quantification | Conformal Prediction | Counterfactual Explanations

EDUCATION

MS/MPhil in Computer Science

University of Gujrat, Gujrat, Pakistan

2025–Present

CGPA: 3.92/4.00

- Coursework completed; thesis and research phase.
- Thesis Supervisor: Dr. Naveed Anwer Butt.

BS in Computer Science

University of Sargodha, Gujrat, Pakistan

2020–2024

CGPA: 3.91/4.00

- Gold Medalist.

RESEARCH EXPERIENCE

MS Researcher | Artificial Intelligence for Healthcare

University of Gujrat, Gujrat, Pakistan

2025–Present

- Conducting research in **medical imaging, multimodal vision-language models, Explainable AI, and trustworthy machine learning**.
- Developing reliability-aware clinical decision-support methods using **conformal prediction, uncertainty quantification, counterfactual explanations, and quantitative XAI evaluation**.
- Applying **PyTorch, TensorFlow, BiomedCLIP, PubMedBERT, Grad-CAM, SHAP, and Quantus**.
- Designing rigorous experiments using independent evaluation, ablation studies, multi-seed experiments, and statistical analysis.

College Teaching Intern (CTI)

Govt. Fatima Jinnah Graduate College, Gujrat, Pakistan

Sep 2026–Present

- Supporting undergraduate students in computer science and programming-related coursework through teaching and academic guidance.
- Assisting students with programming concepts, laboratory work, assignments, and technical problem-solving.
- Providing academic and technical support for computer science and AI-related projects.

Data Analysis & AI Trainee

University of Gujrat, Gujrat, Pakistan

Sep 2025–Dec 2025

- Completed practical training in **data analysis, preprocessing, exploratory data analysis, and visualization**.
- Studied **Natural Language Processing (NLP)** and **Large Language Models (LLMs)** and their applications in AI.
- Applied Python-based techniques for data manipulation, analysis, visualization, and interpretation.

PUBLICATIONS & MANUSCRIPTS

Under Review

1. Butt, N. A., **Nawaz, H.**, Zain Ul Abideen, R. M., Hernandez Cuello, A. M., Azpiazu Saiz, J., de la Torre Diez, I., & Ashra, I. *Towards Reliable Radiology Decision Support: A Composite Trust Score for Conformalized Vision-Language Models with Counterfactual Latent Explanations*. *Frontiers in Physiology*. Under review, 2026.
2. **Nawaz, H.**, Butt, N. A., Zain Ul Abideen, R. M., Rodriguez Velasco, C. L., Leon Brizuela, L., de la Torre Diez, I., & Ashraf, I. *Subject-Wise Validation for Trustworthy AI: An Explainable Transfer Learning Framework for Alzheimer's Disease Classification*. *Frontiers in Neuroscience* (Decision Neuroscience). Under review, 2026. Manuscript ID: 1970598.

In Preparation

3. **Nawaz, H.**, et al. *Reliable Multimodal Vision-Language Models for Medical Imaging Under Distribution Shift: Uncertainty, Calibration, and Explanation Robustness*. Manuscript in preparation, 2026.

SELECTED RESEARCH PROJECTS

Systematic Mapping Study of Explainable AI Robustness

2026

- Conducted a **PRISMA-based systematic mapping study** covering a 173-paper corpus to investigate the robustness and reliability of Explainable AI (XAI) methods.
- Synthesized research trends, evaluation practices, robustness factors, and methodological gaps to identify directions for reliable and reproducible XAI research.

AI-Based Oral Cancer Detection

Python, CNNs | 2026

- Developed a **CNN-based medical image classification system** for oral cancer detection using clinical and histopathology images.
- Performed image preprocessing, model training, and evaluation for computer-aided cancer detection.

Alzheimer’s Disease Classification

Deep Learning, XAI | 2025–2026

- Investigated deep-learning-based Alzheimer’s disease classification with emphasis on **subject-wise validation and explainability**.
- Applied explainability methods including **Grad-CAM, SHAP, and quantitative explanation evaluation**.

TECHNICAL SKILLS

Programming	Python, C/C++, Java, JavaScript, MATLAB, PHP
Deep Learning	PyTorch, TensorFlow, CNNs, transfer learning, fine-tuning
Medical AI	Medical image classification, preprocessing, model evaluation
Trustworthy AI	Explainable AI, Grad-CAM, SHAP, saliency maps, occlusion sensitivity, Quantus, uncertainty quantification, conformal prediction, counterfactual explanations
Multimodal AI	BiomedCLIP, PubMedBERT, vision–language models, multimodal fusion
Data Science	Pandas, NumPy, Matplotlib, SciPy, EDA, statistical evaluation
Development	React.js, HTML, CSS, Tailwind CSS, Flask, FastAPI, REST APIs, Streamlit
Computing & Tools	RunPod, NVIDIA GPU computing, Google Colab, Jupyter, Git, GitHub, Linux, L ^A T _E X

CERTIFICATIONS

- **Ethics, Integrity and Digital Transformation: Navigating Research in the AI Age** — Elsevier, 2026
- **Introduction to AI Agents** — DataCamp, 2026
- **AI-Assisted Coding for Developers** — DataCamp, 2026
- **Data Is Everywhere** — Google, 2025

REFERENCES

Dr. Naveed Anwer Butt — Thesis Supervisor & Course Coordinator
Department of Computer Science, University of Gujrat, Gujrat, Pakistan
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Chaudhary Irfan Ullah — Assistant Professor & Head, Department of Computer Science
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